Module 5

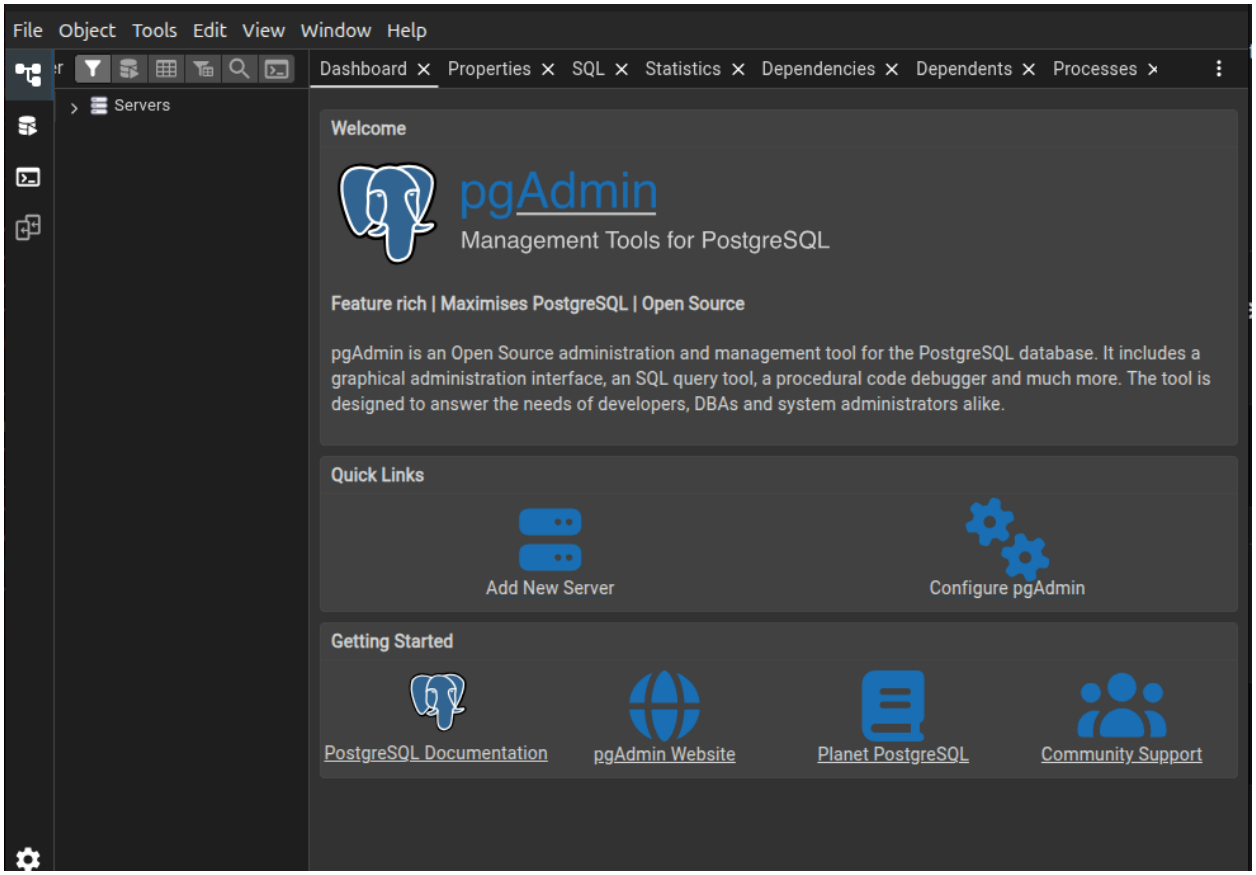
Exercise 1(Intro to PostgreSQL)

1. Setup Postgres Install any latest stable version. Run using the terminal and create a database and table in that.

A1 .) PostgreSQL CLI

```
running bootstrap script ... ok
performing post-bootstrap initialization ... ok
syncing data to disk ... ok
Setting up postgresql-contrib (16+257build1.1) ...
Setting up postgresql (16+257build1.1) ...
Processing triggers for man-db (2.12.0-4build2) ...
Processing triggers for libc-bin (2.39-0ubuntu8.7) ...
dhruv-pandey@TTNPL-dhruv:~ $psql --version
psql (PostgreSQL) 16.11 (Ubuntu 16.11-0ubuntu0.24.04.1)
dhruv-pandey@TTNPL-dhruv:~ $
```

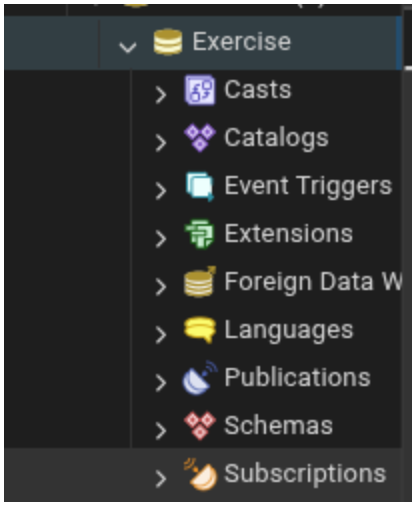
Pgadmin



Create Database& Table

```
postgres=# create database Exercise
```

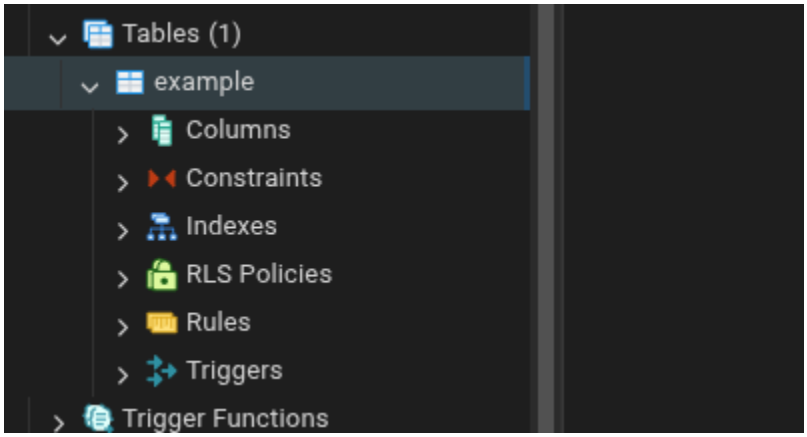
Output



Query Query History

```
create Table example(
id Int,
name text,
subject text
);
```

Output



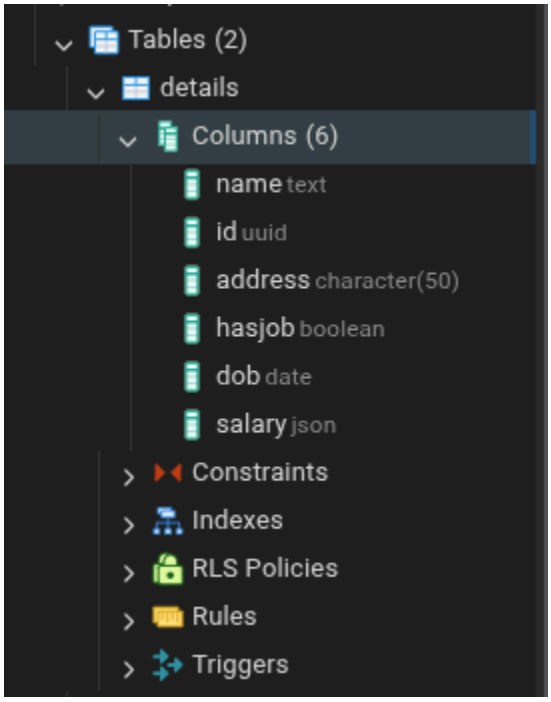
2. Create a table having all the DataTypes in PostgreSQL

A2.) Creating table

Query Query History

```
create table Details(
name text,
id uuid,
address char(50),
hasJob boolean,
DOB date,
salary json
);
```

Output



3. Create a table with primary key constraints

A3.) Creating a table with primary key

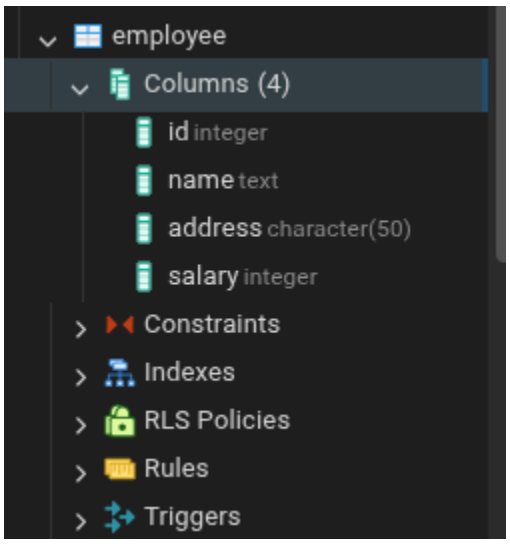
Query Query History

```sql
create table Employee(
id int primary key,
name text,
address char(50),
salary int
);
```

Query Query History

```sql
insert into Employee (id,name,address,salary) values (1,'Dhruv','Delhi',15000),
(2,'Harsh','Delhi',15000),
(3,'Aman','Janakpuri',20000),
(4,'Rohit','Mumbai',50000);
```

Output



Data Output Messages Notifications

	id [PK] integer	name text	address character (50)	salary integer
1	1	Dhruv	Delhi	15000
2	2	Harsh	Delhi	15000
3	3	Aman	Janakpuri ...	20000
4	4	Rohit	Mumbai ...	50000

Error

Query Query History

```
insert into Employee (id,name,address,salary) values
(4,'Krish','Mumbai',50000);
```

Data Output Messages Notifications

```
ERROR:  duplicate key value violates unique constraint "employee_pkey"
Key (id)=(4) already exists.

SQL state: 23505
Detail: Key (id)=(4) already exists.
```

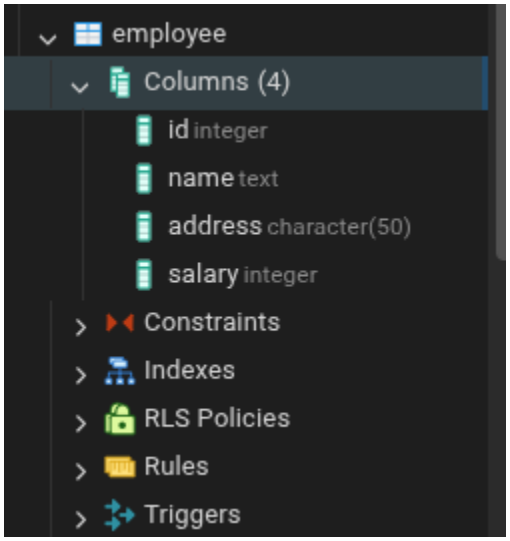
4. Run all basic CRUD queries on the table

A4.)

Create(C)

Query Query History

```
create table Employee(
id int primary key,
name text,
address char(50),
salary int
);
```



Read(R)

Query Query History

```
select * from employee;
```

Data Output Messages Notifications

	id [PK] integer	name text	address character (50)	salary integer
1	1	Dhruv	Delhi	15000
2	2	Harsh	Delhi	15000
3	3	Aman	Janakpuri ...	20000
4	4	Rohit	Mumbai ...	50000

Update(U)

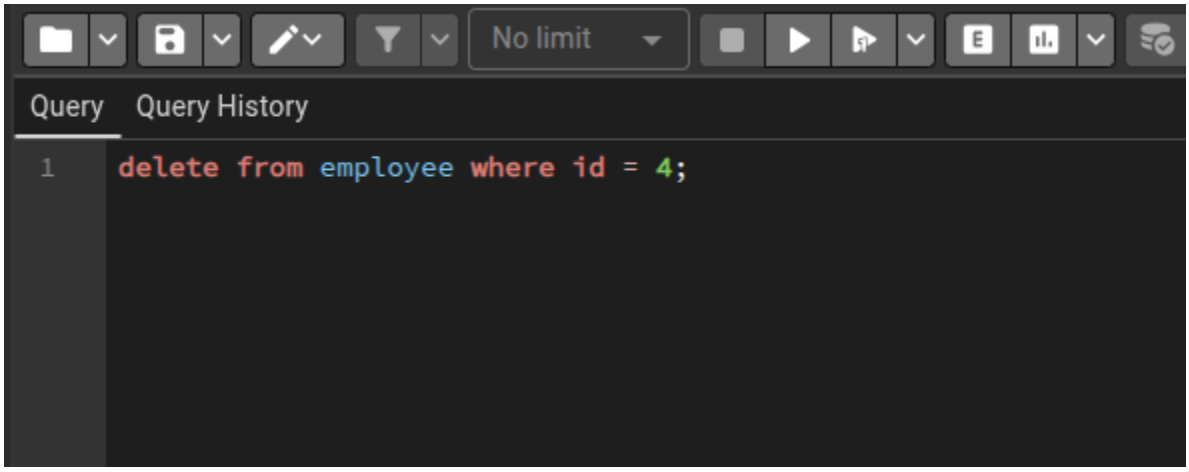
Query Query History

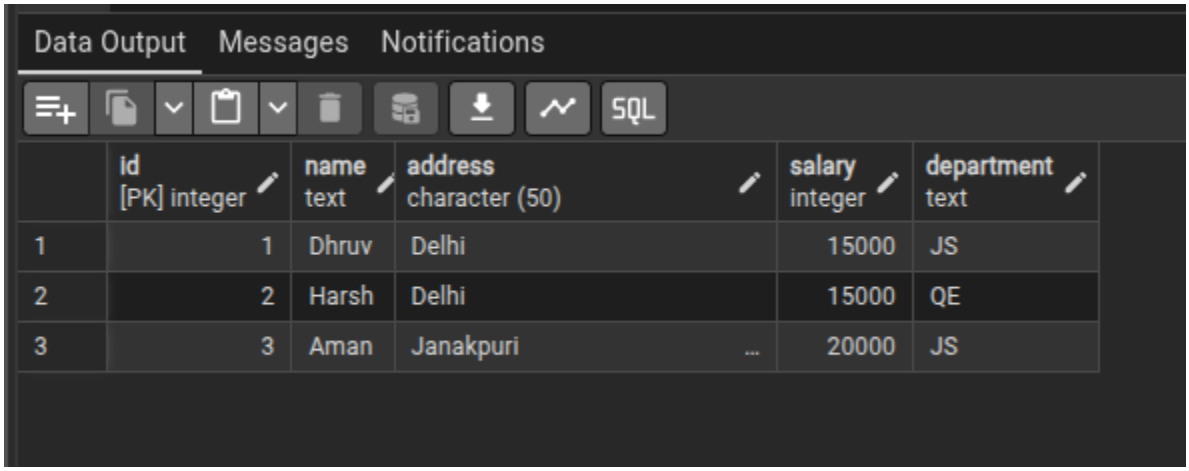
```
update employee set department='JS' where id=1;
update employee set department='QE' where id=2;
update employee set department='JS' where id=3;
update employee set department='JAVA' where id=4;
```

Graph Visualiser

	id [PK] integer	name text	address character	salary integer	department text
1	1	Dhruv	Delhi	15000	JS
2	2	Harsh	Delhi	15000	QE
3	3	Aman	Janakpuri ...	20000	JS
4	4	Rohit	Mumbai ...	50000	JAVA

Delete(D)





	id [PK] integer	name text	address character (50)	salary integer	department text
1	1	Dhruv	Delhi	15000	JS
2	2	Harsh	Delhi	15000	QE
3	3	Aman	Janakpuri	20000	JS

5. Implement GROUP BY, ORDER BY, HAVING in filters.

A5.)

GroupBy

Query Query History

```
select department, sum(salary) from employee group by department;
```

Data Output Messages Notifications

	department text	sum bigint
1	JS	35000
2	QE	15000
3	JAVA	50000

OrderBy

No limit

Query Query History

```
select * from employee order by salary desc;
```

Data Output Messages Notifications

	id [PK] integer	name text	address character (50)	salary integer	department text
1	4	Rohit	Mumbai	50000	JAVA
2	3	Aman	Janakpuri	20000	JS
3	1	Dhruv	Delhi	15000	JS
4	2	Harsh	Delhi	15000	QE

Having

No limit

Query Query History

```
select department, sum(salary) from employee group by department having sum(salary)>15000;
```

Data Output Messages Notifications

	department text	sum bigint
1	JS	35000
2	JAVA	50000